CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 442 Practice

12/01/2023 · 1.5 hours · 25 marks

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01 Questions

Q2 FILL IN + TRUE / FALSE · 2.5 + 1.5 M

a) Fill in the blanks

• InsertItem to dictionary takes _____ time.

• Bottom-up heap construction runs in _____ time.

• In the binary search tree: Let u, v, and w be three nodes such that u is in the left subtree of v and w is in the right subtree of v. Therefore we have: key(u) ≤ _____ ≤ key(w).

• The space used by a Queue with a Singly Linked list is _____

• FindElement in binary tree takes _____

b) Put T or F for each of the following statements.

( ) In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.

( ) Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.

( ) The total of primitive operations represents the Worst case of an algorithm.

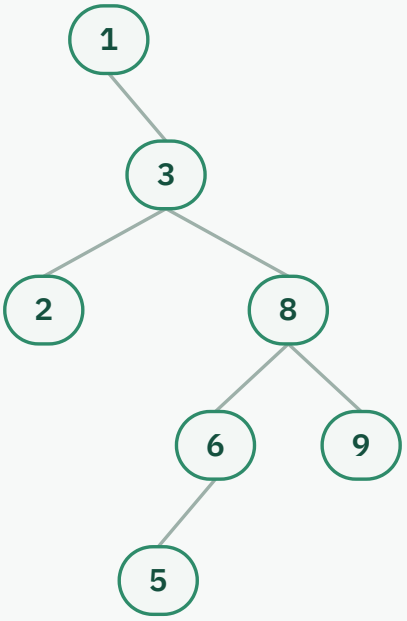
Q3 HEAPS + HASH FUNCTION · 4 + 2 M

a) what is the objective of Upheap, Downheap algorithms? when to use? then what time required to run UpHeap and DownHeap?

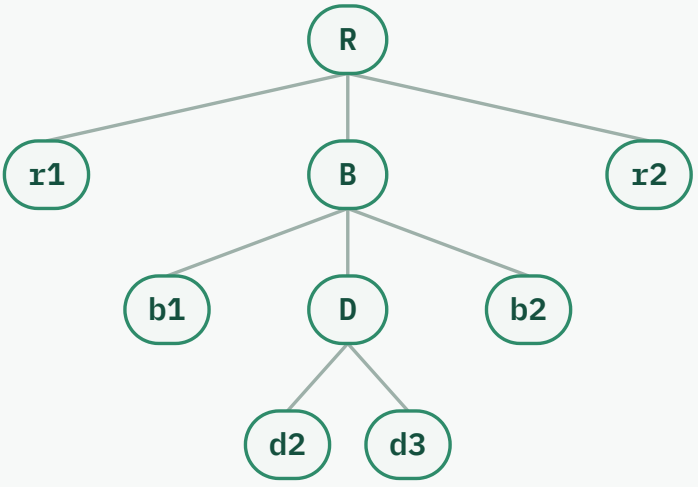
b) When you can say that the hash function is "good"?

Q4 TREES · 2 + 4 M

a) Remove the node 3 from the following binary tree, then redraw the tree after deletion.



b) Design the Linked Data Structure that represents the following Tree structure.



Q5 COLLISION HANDLING · 3 M

Discuss all available methods for handling collisions in hash tables.

Q6 COUNTING OPERATIONS · 2 M

Count the total primitive operations of the following algorithm.

```
Algorithm arrayMax(A, n)
   currentMax ← A[0]
   Temp ← A[0]
   for i ← 1 to n − 1 do
      if A[i] > currentMax then
         currentMax ← A[i]
   { increment counter i }
   return currentMax
```